Introduction

Generation, Transmission and Distribution systems are the main components of an electric power system. Generating stations and distribution systems are connected through transmission lines. Normally, transmission lines imply the bulk transfer of power by high-voltage links between main load centers. On the other hand, distribution system is mainly responsible for the conveyance of this power to the consumers by means of low voltage networks.

Generating stations and a distribution system are connected through transmission lines, which also connect one power system (grid, area) to another. A distribution system connects all the loads in a particular area to the transmission lines.

For economically and technical reasons, individual power systems are organized in the form of electrically connected areas or regional grids (also called power pools). Each area or regional grid are operated technically and economically independently, but these are eventually interconnected to form a "national grid" (which may even form an International grid) so that each area is contractually tied to other area in respect to certain generation and scheduling features. India is now heading for a national grid.

Electric power is generated at a low voltage of 11 to 25 KV which then is stepped up to transmission levels in the range of 66 to 765 KV. As the transmission capability of a line is proportional to the square of its voltage, research is continuously being carried out to raise the transmission voltages. In India, several 400 KV lines are already in operation. Several 765 KV lines have been built so far in India. For very long distances (over 600 Km), it is economical to transmit bulk power by DC transmission.

The first step down of voltage from transmission level is at the bulk power substation, where the reduction is to a range of 33 to 132 KV, depending on the transmission line voltage. Some industries may require power at these voltage levels. This step down is from the transmission and grid level to sub transmission level.

The next step down in voltage is at the distribution substation. Normally, two distribution voltage levels are employed:

1. The primary or feeder voltage (11KV)
2. The secondary or consumer voltage (415 V three phase / 240 V single phase)

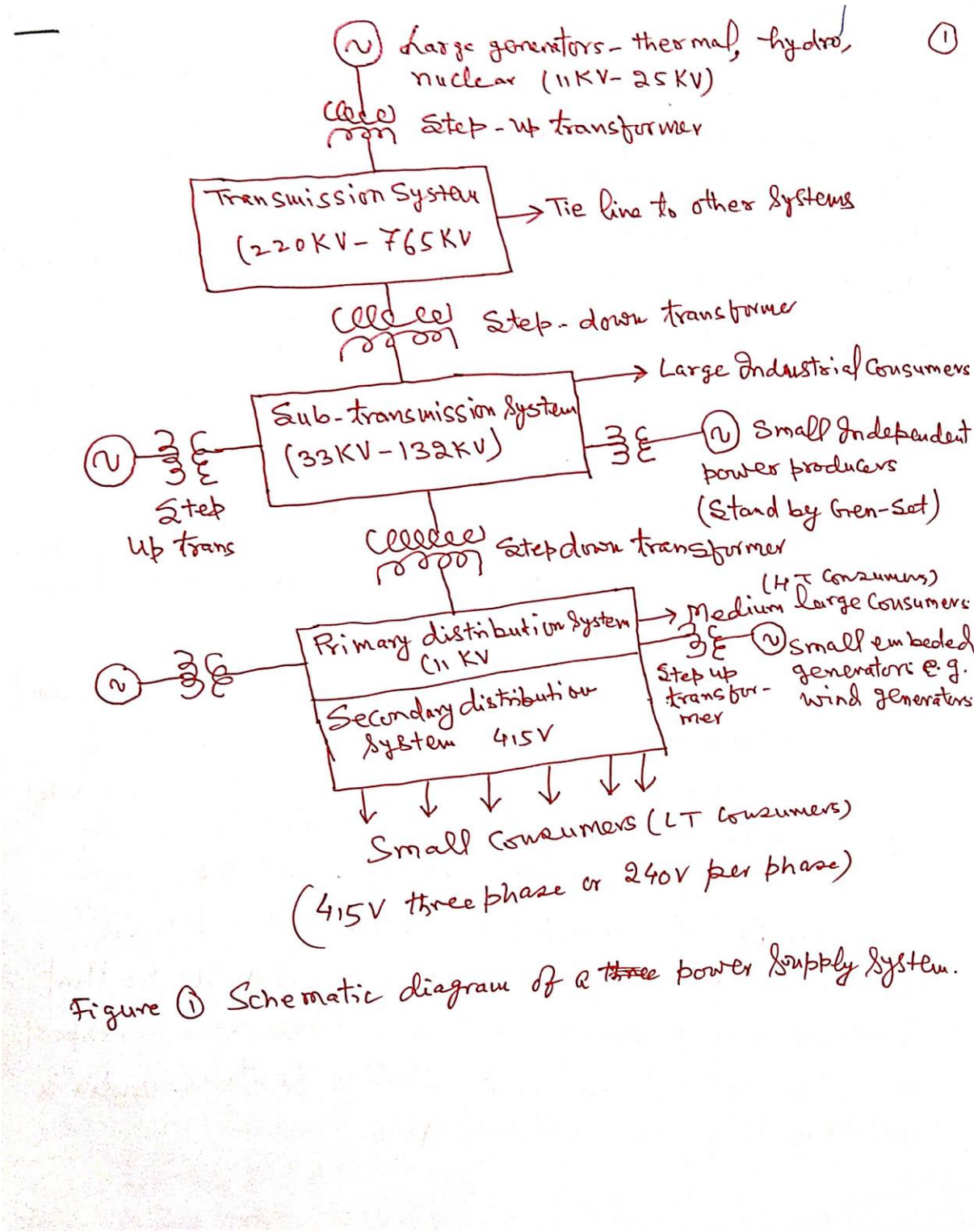


Figure ① Schematic diagram of a ~~three~~ power Supply System.

The distribution system, fed from the distribution transformer stations, supplies power to the domestic or industrial and commercial consumers.

Thus, the power system operates at various voltage levels separated by transformers.

During the Third Five Year Plan (1961-66) transmission growth took place very rapidly. A significant development in this phase was the emergence of an inter-state grid system. The

country was divided into five regions, each with a regional electricity board, to promote integrated operation of the constituent power systems.

- Northern region- Delhi, Haryana, HP, Punjab, UP, UK, Raj, Laddakh, J&K
- Northeastern region- Tripura, Mizoram, Meghalaya, Nagaland, Manipur, Assam, Arunachal Pradesh
- Eastern region- Bihar, Jharkhand, Odisha, Sikkim, Bengal
- Western region- Chhattisgarh, Gujarat, Goa, MP, Maharashtra
- Southern region- AP, TN, Karnataka, Pondicherry, Telangana

Power Stations and Sub-stations

Electrical network comprises the following regions:

(1) Generating stations (2) Transmission systems and (3) Distribution Systems

In these entire regions, the power flow of electrical energy takes place through Electrical substations. An electrical sub-station is an assemblage of electrical components including bus bars, switchgears, power transformers, auxiliaries, etc. The substations are located in generating stations, transmission and distribution systems and in consumers premises. Generally, an electrical substation consists of a number of incoming and outgoing circuits connected to common bus bar system. Bus bars are conducting bars of large current carrying capacity to which a number of incoming and/or outgoing circuits are connected. Each circuit has certain electrical components such as circuit breakers, isolators, earthing switches, C.T., P.T. etc. These components are connected in a different sequence such that a circuit can be switched off during normal operation by manual command. Electrical energy is generally generated in large hydro, thermal and nuclear power stations but the power stations are located far away from the load centers. Hence, large and long transmission network are used to carry the power from these power stations to substations in the load centers. Several electrical equipments are used for proper transmission and distribution of the generated power. The transmission system of an area is popularly known as ‘grid”. Different grids are interconnected through tie lines to form a regional grid. Several regional grids are further interconnected to form a national grid. Each grid operates independently, though power can be exchanged between grids also.

Non-conventional generations of electric energy (Renewable energy sources) are now being actively considered. This serves mainly as a local source of electric power.